SEEING CHANGE FROM ORBIT · MEETING MINUTES

Meeting 6: Mangrove Model Comparison

Date and time	Mon 26 Oct 2026, 1:30–2:30 pm
Type	Supervisor
Location	Online (MS Teams)
Attendees	Dr. Panos, Khaled, Zain, Ayesha, Tehsin, Zayed

Agenda

Agenda sent beforehand: the model-comparison document.

1. More UAE sites
2. Reference data and labelling
3. Model results and next steps

Member updates

Who	Update
Ayesha	Presented the model-comparison document
Zayed	Shared his updates (not detailed in the record)
Tehsin	Extended registration work to more UAE mangrove locations
Zain	Multispectral Sentinel-2 exports
Tehsin	Reviewed the GMW v4.1 methodology: time-series layers from Landsat and Sentinel-2, multiple baseline years, classification by an XGBoost model (so GMW is machine-generated, not hand-labelled)
Tehsin	Reviewed how GMW handles tide, cloud, seasonal effects and class imbalance when comparing aligned regions over time; searching for a more reliable ground-truth source

Discussion

1. More UAE sites

- Ayesha asked whether to use more UAE mangrove sites than the three Abu Dhabi sites. Dr. Panos: more data helps training, as long as it is labelled and can be prepared for the model.

2. Reference data and labelling

- GMW labels turn out to come from an XGBoost classifier, not from people, so its accuracy is uncertain.
- Ayesha can contact DMT (where she interned) about mangrove data for educational use, possibly under an NDA.
- In parallel: hand-label a small sample (e.g. five images from different sites) in mixed pairs of team members, without looking at GMW, then compare with GMW. If close, assume GMW's accuracy carries over to the rest.

- How to handle disagreements between labelers is not decided: keep only agreed areas (conservative) or the union (larger ground truth).

3. Model results and next steps

- Cross-validation on six Abu Dhabi sites (locked test sites untouched): U-Net IoU 0.71, random forest 0.70, frozen DINOv3 0.67, NDVI + MNDWI rule 0.60. Two GMW releases agree only at IoU 0.47–0.83, so ~0.70 is near the label ceiling. Jubail is the hardest site (~0.55).
- Next: fine-tune DINOv3 with LoRA at higher input resolution, then pick the final model.
- Dr. Panos answered from the document: choose the final model from cross-validation only; if two models tie, pick the simpler one. He needs to check what data DINOv3 was trained on.
- Dr. Panos: very good progress, considering midterms.

Decisions

- Final model is chosen from cross-validation only; the locked test sites are scored once.
- If two models tie, choose the simpler one.
- Fine-tune DINOv3 with LoRA before the final choice.
- Check GMW quality with a small, paired hand-labelled sample and a DMT data request.

Action items

Who	What	Due
Ayesha	Investigate GMW methodology and classification accuracy	Not set
Ayesha	Contact DMT about sharing mangrove data for educational use	Not set
Team	Fine-tune DINOv3 with LoRA	Not set
Dr. Panos	Check what data DINOv3 was trained on	Not set
Team	Hand-label a small sample in mixed pairs and compare with GMW	Not set
Team	Decide how to resolve labeler disagreements	Not set

Next meeting

Mon 2 Nov 2026, 1:30–2:30 pm, Online (MS Teams)